

being fed into YOLOv5 for object detection. It demonstrates how different modules respond to the same input at different stages of the network. Module 1 produces a PEEK map that highlights fine-grained details of the satellite. Module 19 creates a richer and more comprehensive representation capturing the full areas of interest, although, it can also be falsely triggered by background noise. In Chapter 3, we argue that Module 19 is more informative qualitatively as it focuses on the satellite. However, we lack an approach to evaluate the usefulness of the visualizations.

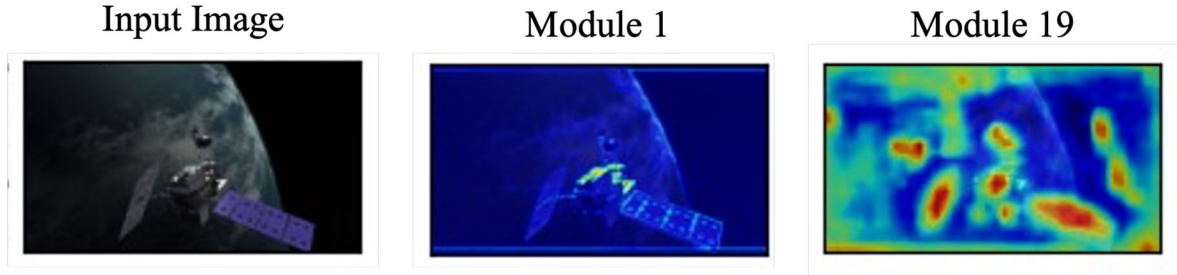


Figure 4.1: A comparison of PEEK for Module 1 and Module 7 in YOLOv5 with satellite data seen in Chapter 3

However, if we were to break down the way we as humans evaluate the PEEK maps in a neural network, we tend to intuitively judge them based on how informative, detailed, and meaningful the representation appears compared to the goal (detecting the satellite components here). In practice, this judgment often corresponds to how much variation is visible in these visualizations. A PEEK map that highlights many distinct patterns or areas of interest across the input data, like the different sections of the satellite, it is likely perceived to be informative. On the other hand, maps with low variations may appear uninformative, suggesting that the layer or module didn't extract meaningful information.

This human intuition aligns closely with the idea of variance of PEEK maps. Variance measures how much individual elements deviate from the mean. If we see areas of high and low entropy distributed meaningfully across the PEEK map, it indicates the

module is capturing a diverse range of patterns, resulting in high variance. Conversely, low variance suggests the module’s entropy is relatively uniform, meaning it is contributing little unique or varied information to the prediction. For example, Module 19 highlights multiple regions, including the objects on the satellite. This suggests that the module has identified diverse and important features, each element in the PEEK map reflects varying levels of activation, resulting in high variance.

In order to explore if we can connect human intuition of the qualitative aspects of the image, we calculate the variance of the PEEK maps.

4.3.2 Variance of PEEK: Mathematical Explanation

The variance measures the spread or dispersion of values in the PEEK map. To compute the variance of PEEK at convolutional layer ℓ , we follow these steps:

First, extract the activations at layer ℓ , $A^\ell \in \mathbb{R}^{l \times w \times c}$. Then, adjust the values to ensure all elements are non-negative by adding the minimum intensity across all channels:

$$A_{ijk}^{\ell, \text{pos}} = A_{ijk}^\ell + \left| \min_{\{i \leq l, j \leq w\}} A_{ijk}^\ell \right|, \quad (4.1)$$

where A_{ijk}^ℓ represents the activation at position (i, j) in the k th channel. Next, we compute the pseudo-entropy for each element using $g(x) = -x \ln x$ and sum depthwise at each position (i, j) across all channels:

$$P_{ij} = - \sum_{k=1}^d g \left(A_{ijk}^{\ell, \text{pos}} \right) \quad (4.2)$$

This produces the PEEK map P^ℓ , a 2D matrix where each element P_{ij}^ℓ represents the activation entropy value (or PEEK) for each spatial location. It is then upsampled

to the size of the input images using bilinear interpolation and antialiasing to produce matrix $Q^\ell \in \mathbb{R}^{L \times W}$ where L and W are the length and width of the input image.

The mean μ_ℓ of the PEEK map is:

$$\mu_\ell = \frac{1}{LW} \sum_{i=1}^L \sum_{j=1}^W Q_{ij}^\ell \quad (4.3)$$

where LW is the total number of elements in the matrix. The variance measures the average squared deviation from the mean:

$$\sigma_\ell = \text{Var}(P^\ell) = \frac{1}{LW} \sum_{i=1}^L \sum_{j=1}^W (Q_{ij}^\ell - \mu_\ell)^2 \quad (4.4)$$

This provides insight into how spread out the values in the PEEK map are. A high variance indicates that certain areas or features have much higher activation than others, suggesting the model is focusing on specific regions. Conversely, low variance suggests a more uniform activation across the map.

- High variance: Significant deviations from the mean suggest that certain activations are much stronger, indicating focused attention on key regions.
- Low variance: More uniform activations suggest generalized attention across the input space, potentially indicating under-optimization or lack of feature specificity.

4.3.3 Understanding the Relationship of Variance in PEEK and Model Accuracy

We trained simple convolutional neural networks (CNNs) on CIFAR-10 dataset to investigate the relationship between variance in PEEK maps and a model’s accuracy.

We started with a small architecture containing three convolutional layers, then incrementally increased the number of layers while maintaining the same optimizers, activations, and layer depths in the prior models until we observed a significant drop in performance. For each architecture, we evaluated the variance of PEEK maps across 100 images to assess how variance correlates with accuracy.

The CIFAR-10 dataset is a standard benchmark dataset for image classification tasks containing 60,000 color images distributed across 10 classes, with each class representing a different category, such as airplanes, cars, birds, and dogs. The images are small (32 x 32 pixels), making this dataset ideal for training and evaluating compact models. CIFAR-10 is challenging due to the classes containing various visual patterns, requiring the models to learn diverse features.

For each architecture, we monitored accuracy on the validation set. When we observed a significant accuracy drop, we concluded that further increasing the number of layers was not beneficial. Each architecture was evaluated independently to track how changes in layer depth impacted performance.

After training, we computed PEEK maps for 100 randomly selected images from the validation set for each network. Variance of PEEK was calculated for each layer across those images to determine how much the entropy values deviated from the mean values. The hypothesis was that layers or networks with higher variance patterns would encode more informative and varied features, potentially leading to better generalizations and higher accuracy.

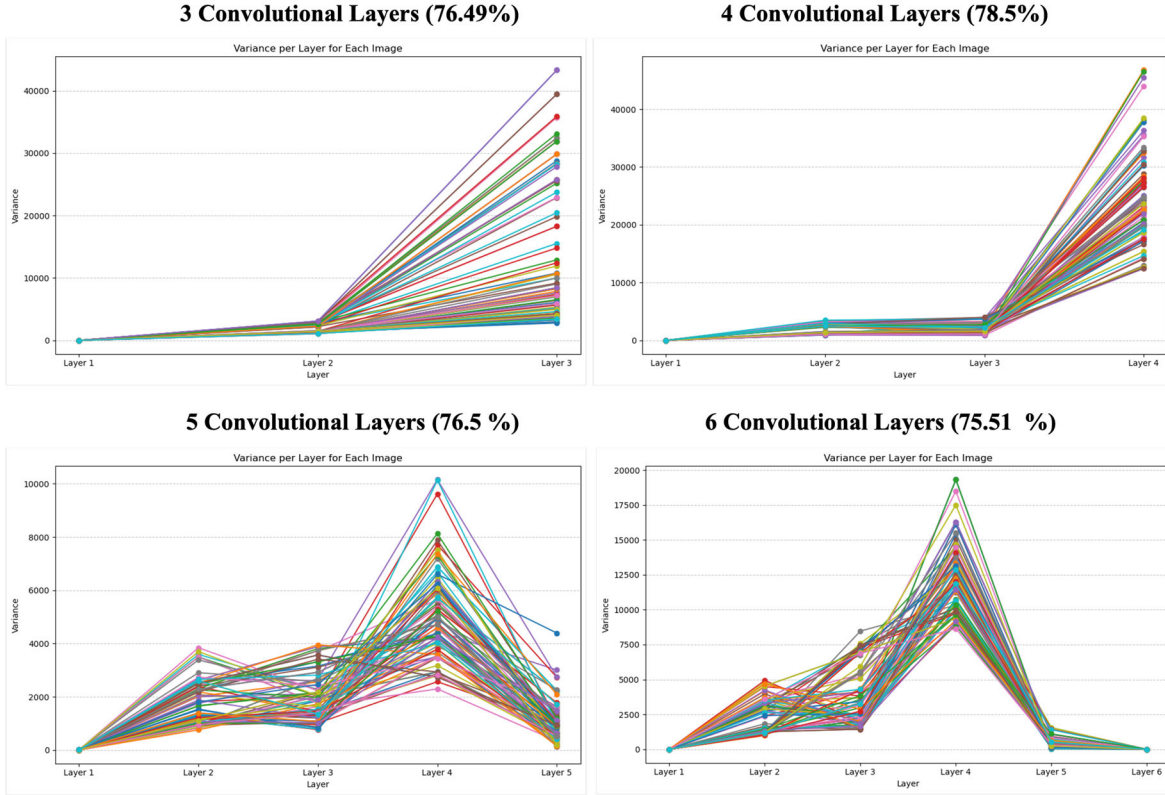


Figure 4.2: This image shows the variance in PEEK maps of 100 images through the layers of 4 simple CNNs trained with incrementally additional layers.

In Figure 4.2 above, it shows the variance trends and accuracy for networks with 3 to 6 convolutional layers. Key findings include:

- **3 Convolutional Layers:** The variance increases sharply at layer 3, suggesting the final convolutional layer plays a significant role in capturing meaningful patterns. However, the overall accuracy of this model is lower compared to deeper models, indicating that the architecture may be too shallow to capture all relevant features.
- **4 Convolutional Layers:** With the inclusion of an additional layer, the variance increases a slightly more gradually across the layers, especially towards the final layer. This model achieves better accuracy, suggesting that the additional layer

helps the model capture more useful representations.

- 5 Convolutional Layers: This model shows a peak and drop in variance at layer 4. The accuracy decreases compared to the 4 layer network, suggesting that the network may be starting to introduce redundancy.
- 6 Convolutional Layers: Despite the additional layers, the overall accuracy drops lower than the 3 convolutional layer model. The variance peaks at intermediate layers, but falls sharply towards later layers, indicating that the deeper network struggles to maintain meaningful activations across all layers.

This experiment showed that networks with higher variance in PEEK maps at critical layers (such as the last convolutional layers) tend to perform better as the layer 4 network had the highest validation accuracy. This suggests that the variance is an indicator of how well the network is capturing useful patterns. Networks with more layers exhibited peaks and drops in variance, indicating that not all layers contribute equally to the final output. This aligns with the need for restructuring-type pruning, where layers with lower variance are pruned to reduce model complexity without sacrificing performance.

4.3.4 Relative Variance Contribution (RVC)

The concept of Relative Variance Contribution (RVC) allows us to quantify the proportion of variance contributed by each layer in a neural network. Given that each layer ℓ produces its own variance σ_ℓ^2 from its respective PEEK map, we can calculate both the total variance across all layers and the proportion contributed by each individual layer.

The **total variance** for all layers is:

$$\text{Total Variance} = \sum_{\ell} \sigma_{\ell}^2 \quad (4.5)$$

where σ_{ℓ}^2 is the variance of the PEEK map for the ℓ th layer.

The **Relative Variance Contribution (RVC)** for each layer ℓ is defined as the percentage of the total variance that the variance of the ℓ th layer contributes. This is given by:

$$\text{RVC}_{\ell} = \left(\frac{\sigma_{\ell}^2}{\text{Total Variance}} \right) \times 100 \quad (4.6)$$

This formulation ensures that the sum of all RVC values across the layers equals 100%, representing the total distribution of variance contributions across the network.

This approach ensures a precise understanding of how much each layer’s PEEK map contributes to the overall variance. These insights can be used for network pruning, optimization, and improving model interpretability by identifying key layers that influence the variance the most.

4.4 Results

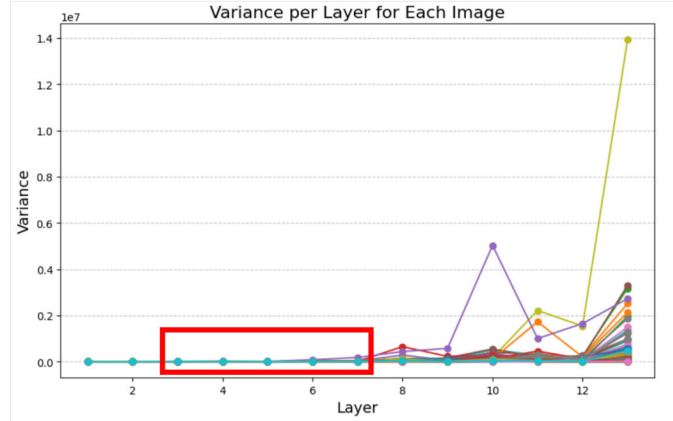
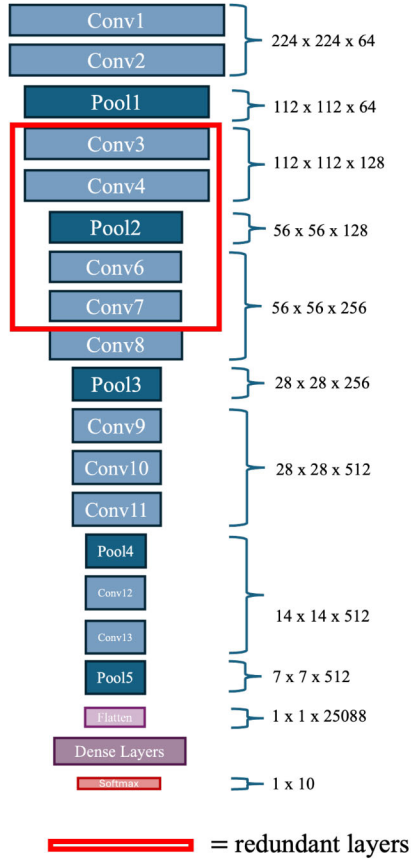
4.4.1 PEEK Pruning VGG-16

In this section, we explore how structural pruning was applied to the VGG-17 architecture using insights derived from the RVC analysis. Structural pruning removes entire layers or modules, reducing the model’s complexity while aiming to maintain performance. This approach is particularly beneficial for edge deployments, where the computational resources are limited.

We trained a VGG-16 model from scratch on the Imagenette dataset, a lightweight subset of ImageNet. It contains 10 easily recognizable classes, such as dogs, flowers, and buildings, serving as a useful benchmark for experimentation. After training, we fed 100 images through the trained model to compute the PEEK maps for each layer, from which we calculated the variance of. Using these variances, we derived RVC metrics for each layer, revealing how much each layer contributed to the network’s overall PEEK variance.

From the RVC analysis, we identified that some layers in the middle of the network (convolutional 3-7) contributed relatively low variance, under 2% of the total variance in each layer, compared to the later layers as seen in Figure 4.3 below.

Baseline VGG-16



Conv Layer #	RVC %
1	0.003943
2	0.160747
3	1.079124
4	0.702691
5	0.699294
6	1.346708
7	1.502884
8	9.894344
9	6.218063
12	7.943786

Figure 4.3: VGG-16 architecture with redundant layers highlighted in red (Conv3–Conv7), identified through PEEK variance analysis. The variance plot (top right) shows low variance across these layers, indicating minimal contribution to performance. The table below lists each layer’s Relative Variance Contribution (RVC), with redundant layers displaying notably low values.

At first glance, removing the earliest layers, such as convolutional layers 1 and 2, may seem like a natural step because they also exhibit low RVC. However, these layers are critical for capturing high-level, generic features such as edges, textures, and simple patterns. These features form the foundation upon which deeper layers build more complex representation. Removing these initial layers would potentially undermine the network’s ability to learn hierarchical features, potentially harming performance across the entire model. Alternatively, deeper layers like Convolutional layers 8-12 tend

to learn task-specific features and complex relationships, making them essential for the final predictions. These layers contribute the most to the model’s variance as indicated by their high RVC values (all about 6%).

With these insights, we focused on pruning Layers 3 through 7 (highlighted in red 4.3). These layers contribute relatively low variance to the overall output, sit between high-level feature extraction in the early layers and the task-specific feature extraction in the later layers, and risk redundancy as they often learn intermediate features that overlap with earlier or later layers.

The goal of this method was to find a way to reduce model complexity while maintaining performance suitable for edge deployment. We initially pruned the intermediate layers identified from the RVC analysis, reduced the overall feature dimensionality to 256 to maintain the original increase in depth, and fine-tuned the model from the baseline weights to restore performance (1.0 Proposed VGG-16 4.4). We then conducted further targeted pruning based on the RVC results of the 1.0 model to assess the impact of layer removal at specific locations and fine-tuned each pruned model independently (1.1 and 1.2 Proposed VGG-16 4.4). The architecture for each of these newly structured models can be seen in Figure 4.4.

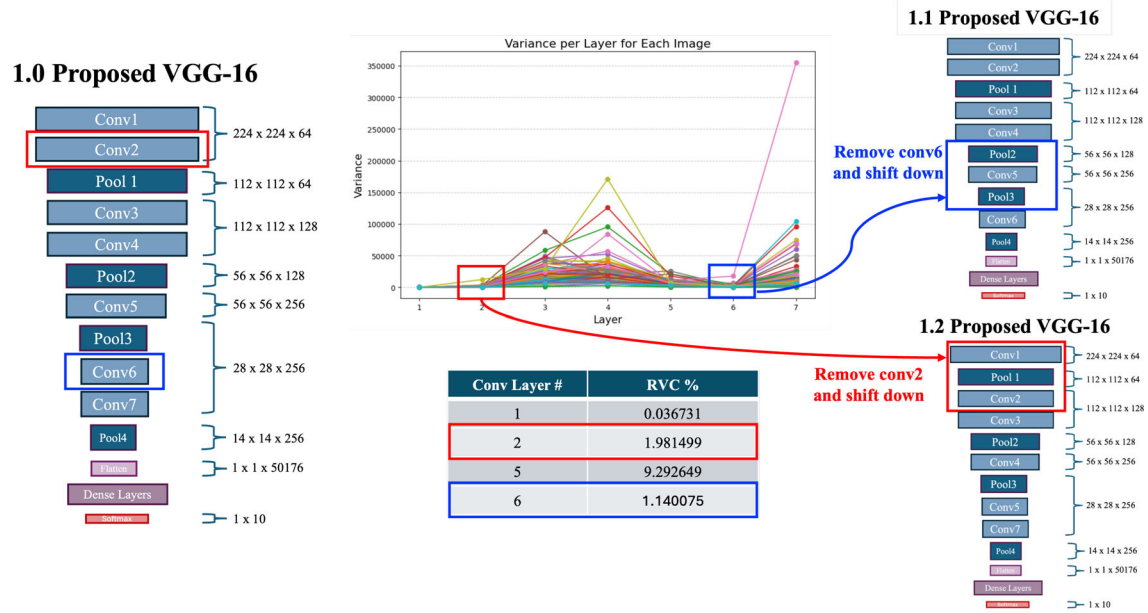


Figure 4.4: VGG-based architecture with proposed pruning modifications based on PEEK variance and Relative Variance Contribution (RVC) analysis. The first pruned architecture 1.0 (left) shows layers with low RVC highlighted in red (Conv2) and blue (Conv6), indicating minimal contribution to model performance. The variance plot (top center) confirms low variance in these layers across images. Pruning options include removing Conv2 in 1.2 (red box) and shifting subsequent layers or removing Conv6 in 1.1 (blue box) and restructuring the network accordingly.

1.0 Proposed VGG-16

This model was the initial structural pruning of layers 3-7 with the addition of reducing the final dimensionality to 256. This first stage of pruning after fine-tuning, achieved a validation accuracy of 90.14% which is slightly higher than the baseline. This demonstrates that pruning the redundant layers based on RVC can lead to a more efficient architecture without sacrificing performance. Removing these layers eliminated unnecessary operation, reducing the model size from 80.89 MB to 55.88MB and GFLOPS from 15.35 to 6.57 4.1, making the network better suited for deployment on low-compute devices.

Model	Validation Accuracy	Parameters	Model Size	GFLOPS
Baseline	90.06%	21,205,834	80.89 MB	15.35
Pruned 1.0	90.14%	14,649,162	55.88 MB	6.57
Pruned 1.1	89.04%	14,059,082	53.63 MB	6.11
Pruned 1.2	87.77%	14,612,234	55.74 MB	4.72

Table 4.1: Model Comparison: Validation Accuracy, Parameters, Model Size, and GFLOPS

After the initial pruning, we calculated the RVC of the pruned network (Model 1.0) to identify further opportunities for reduction. The RVC analysis indicated that convolution layer 6 and convolutional layer 2 had relatively low contributions to the total variance. We proceeded to remove these layers separately to study the impact of pruning at different locations within the network. Both models were adjusted accordingly for continuity and fine-tuned from the baseline model.

1.1 Proposed VGG-16 after fine-tuning, achieved a validation accuracy of 89.04%. Pruning convolutional layer 6 reduced the computation cost further with only a minor impact on accuracy. This model had a slight decrease in GFLOPS but a reduction in model size to 53.63MB 4.1.

However, **1.2 Proposed VGG-16** after fine-tuning, yielded a validation accuracy of 87.77%. While this model had a GFLOPS reduction to 4.72, the model size increase slightly to 55.74MB4.1. This supports the hypothesis that removing earlier layers responsible for capturing generic features hinders the network’s ability to learn effectively.

Overall, this second stage of pruning would only be necessary if you needed to further reduce the size or speed of the model at the risk of accuracy reduction. Pruning earlier layers results in lower GFLOPs (faster inference) but a slightly larger model

size. Layer 2 has less impact on model size since it contributes fewer total parameters, leading to a slightly larger model after pruning. However, since the layer operates at a high spatial resolution (224×224), eliminating it reduces the number of operations significantly. Pruning later layers results in a smaller more accurate model due to the removal of the large channel sizes, but have little impact on speed due to the low resolution.

4.4.2 PEEK Pruning YOLOv5

In Chapter 3, we examined the application of PEEK visualization within a YOLOv5 object detector, specifically trained to identify solar arrays, satellite bodies, antennas, and thrusters. The primary objective of the detector was to eventually adapt and optimize it for deployment on compact, low-power hardware, making it suitable for onboard satellite systems. PEEK-based pruning offers the potential to enable efficient, lightweight, real-time object detection in space environments where resources are limited and performance is critical.

Utilizing the YOLOv5 model trained on unknown satellites from Chapter 3, we passed through a video that was sliced at 30 FPS . We calculated variance of PEEK values for approximately every 14 frames to reach 100 images with no overlap.

The variance of PEEK plot for the YOLOv5 model in Figure 4.5 provides insights into the distribution of the variance across the layers based on multiple input images. This plot shows that Module 0 contains the highest variance across all images, indicating that it plays a critical role in capturing diverse information, such as edges, textures, or simple patterns. This suggests that the model is strongly relying on these features across images for predicting the different objects and potentially not able to extract any complex relationships, only relying on the first module's features to predict.

After module 0, there is a sharp drop in variance. In fact, the variance is nearly

flat across all modules from 8 through 14 (highlighted in red in Figure 4.5). This may indicate that these layers introduce redundancy or are not extracting meaningful diverse features. The flat constant trend suggests that these modules are underutilized, making them potential candidates for pruning. However, modules 15, 18, and 19 contain distinct spikes in variance, which correspond to critical points in the model where the complex features are aggregated and passed to the detection heads.

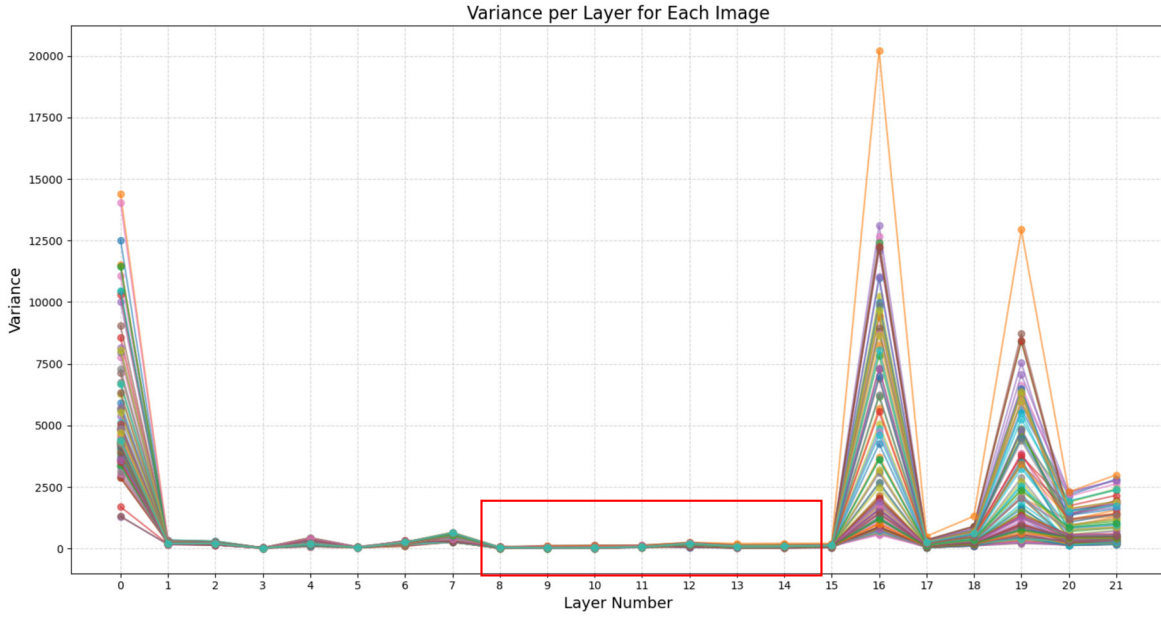


Figure 4.5: Variance per layer for each image across a YOLOv5 object detection network, highlighting layers with distinct peaks in variance.

To confirm the impact of each of the layers, we calculated the RVC through YOLOv5. This showed that 38.33% of the models variance comes from Module 0, making it critical for the network’s performance. However, modules 3 through 14 have very low variance contribution seen in Table 4.2.

Under 1%		1-5%		Over 5%	
Layer	RVC (%)	Layer	RVC (%)	Layer	RVC (%)
3	0.178	1	1.873	0	38.331
5	0.357	2	1.709	16	20.964
8	0.319	4	1.819	19	13.960
9	0.230	6	1.554	12	0.951
10	0.257	7	3.343	15	0.716
11	0.586	18	2.161		
13	0.397	20	4.084		
14	0.435	21	4.924		
17	0.854				

Table 4.2: Categorized Normalized RVC Summary for Each Layer Across All Images

By allowing the RVC values to guide the way we prune the model, initial attempts will focus on modules with $RVC < 1\%$ for removal. Modules 3, 5, 8, 9, 10, 13, and 17 contribute minimally to the network’s variance, therefore, removing them is unlikely to significantly affect the model’s accuracy. However, due to the complex structure of YOLOv5, additional layers will need to be moved and reconnected. The first step would be to choose the modules to remove and then find the best path. This may include removing any concatenation points between redundant layers, or removing the paths entirely.

Pruning these layers will reduce the number of parameters dramatically, making the model lighter and more suitable for edge deployment. Since the modules targeted for removal contribute little to the model, it is expected to maintain or improve accuracy after fine-tuning.

4.5 Conclusion

In this chapter, we explored how PEEK-based variance analysis can guide the process of pruning neural networks, balancing speed, accuracy, and efficiency for edge deployment. The primary objective was to create a framework for reducing model size and computational cost without compromising performance, addressing the real-world challenges of deploying deep learning models on constrained devices like NVIDIA Jetsons, Raspberry Pi boards, and microcontrollers.

Using variance in PEEK maps and Relative Variance Contribution (RVC) as metrics, we identified layers that contributed redundant or minimal information to the network’s performance. Through this structured pruning, we reduced parameters, GFLOPs, and model size, creating models that are optimized for real-time operation in resource-constrained environments such as satellites or autonomous medical systems.

This work provides a foundation for further research on variance-based pruning techniques. Future directions include automating the pruning process using PEEK variance and RVC metrics to dynamically adjust architectures during training. Additionally, we would like to further explore the trade-offs between pruning depth and model accuracy, with a focus on different network architectures, including transformers, LLMs, and generative models. Finally, this framework will be applied to real-world edge deployment scenarios, such as drones and satellite systems. The ability to prune and restructure deep networks systematically using variance metrics is a step toward more scalable and sustainable AI models that can thrive beyond cloud infrastructure and operate independently in aerospace, medical, and military applications.